AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII EXAMINATION

APRIL/ MAY 2017

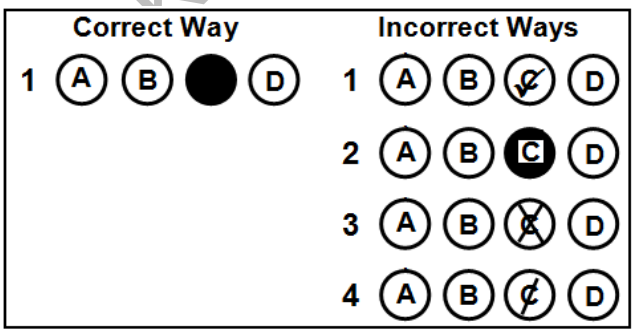
Mathematics Paper I

Time: 45 minutes Marks: 35

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 35 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a scientific calculator if you wish.

1. If $f(x) = ax + b$, then $f^{-1}(x)$ is equal to

   A. $-ax - b$
   B. $\frac{x-b}{a}$
   C. $\frac{x-a}{b}$
   D. $\frac{a}{x-b}$

2. $\lim_{x \to 0}\left(\frac{\sin x + \cos x}{\sin x - \cos x}\right)$ is equal to

   A. -1
   B. 0
   C. 1
   D. ∞

3. If $x \in R - \{2\}$, then the range of $y = -|x-2|$ will be the set of

   A. real numbers.
   B. positive real numbers.
   C. real numbers except 2.
   D. negative real numbers.

4. The domain of $g(x) = \frac{2x-3}{x^2 - a^2}$ is

   A. $R - \{a, -a\}$
   B. $R - \{a^2, -a^2\}$
   C. $R - \{a, -a, 3\}$
   D. $R - \{a^2, -a^2, 3\}$

5. For $y = x^2$, when $x = 2$ and $\delta x = 0.1$, the value of δy is

   A. 0.04
   B. 0.41
   C. 0.42
   D. 8.4

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6. If a function $g(x)$ is said to be continuous at b, then which of the following conditions is/ are correct?

I. $g(b)$ is defined

II. $\lim\limits_{x \to b} g(x)$ exists

III. $\lim\limits_{x \to b} g(x) = g(b)$

A. II only
B. III only
C. I and II
D. I, II and III

7. If $w = \sin x$ and $t = 2x$, then $\frac{dw}{dt}$ is equal to

A. $-2\cos x$
B. $2\cos x$
C. $-\frac{\cos x}{2}$
D. $\frac{\cos x}{2}$

8. If $y = \ln e^{2x}$, then $\frac{dy}{dx}$ is equal to

A. 0
B. 2
C. $2e^{2x}$
D. $\frac{1}{e^{2x}}$

9. If the derivative of a curve y is $\frac{dy}{dx} = 8 - \frac{1}{x^3}$, then x-coordinate of the stationary point is

A. $-\frac{1}{2}$
B. $\frac{1}{2}$
C. 2
D. 8

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10. The slope of the tangent to the curve $y = \frac{7}{4}x + \frac{3}{x}$ at $x = 2$ is

A. 0
B. 1
C. $\frac{5}{2}$
D. $\frac{3}{4}$

11. Which of the following is a proper fraction?

A. $\frac{x(x-1)}{x^2+2}$
B. $\frac{x^2+3x+2}{(x+2)^2}$
C. $\frac{(x-1)(x^2-3)}{x^2+2x+3}$
D. $\frac{x^2+3x+2}{(x^2-1)(x-3)}$

12. $\int \frac{\tan x}{\cos x} dx$ is equal to

A. $\text{cosec}\, x + c$
B. $\sec^2 x + c$
C. $\sec x + c$
D. $\tan x + c$

13. The area bounded by the curve $y = 9 - x^2$ and the x-axis can be found by

A. $\int_0^9 (9 - x^2) dx$
B. $\int_{-9}^9 (9 - x^2) dx$
C. $\int_{-3}^3 (9 - x^2) dx$
D. $\int_0^3 (9 - x^2) dx$

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14. $\int \frac{2x}{x^2} dx$ is equal to

A. $\ln x + c$
B. $\ln x^2 + c$
C. $2x \ln x + c$
D. $2x \ln x^2 + c$

15. $\int (x+b)^{-\frac{1}{2}} dx$ is equal to

A. $2\sqrt{x+b} + c$
B. $-2\sqrt{x+b} + c$
C. $2b\sqrt{x+b} + c$
D. $-2b\sqrt{x+b} + c$

16. The velocity of a car is $v = 2t^2$ m/s during time interval t seconds. Its acceleration at $t = 3$ s is

A. 18 m/s^2
B. 12 m/s^2
C. 6 m/s^2
D. 4 m/s^2

17. The equation of a line parallel to $y = 5x + 2$ is

A. $y = 5x + 10$
B. $y = \frac{1}{5}x + 10$
C. $y = -5x - 10$
D. $y = -\frac{1}{5}x - 10$

18. The equation of a line passing through (0, 0) and (3, 2) is

A. $2x + 3y = 0$
B. $2x - 3y = 0$
C. $3x + 2y = 0$
D. $3x - 2y = 0$

19. Which of the following points lies below the line $x - 3y + 2 = 0$?

A. (– 5, –1)
B. (5, –1)
C. (1, 1)
D. (0, 5)

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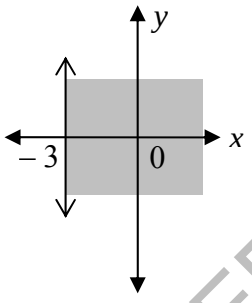
20. The equation of the line that has zero slope and which passes through the point (2, 3) is

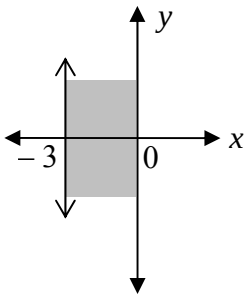
A. $x + 2 = 0$
B. $x - 2 = 0$
C. $y + 3 = 0$
D. $y - 3 = 0$

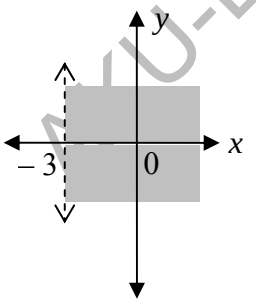
21. The *y*-coordinate of the point dividing the line segment joining the points *A* (2, 0) and *B* (0, 5) internally in the ratio 1 : 3 is

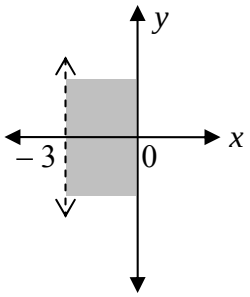
A. $-\frac{15}{2}$
B. $-\frac{5}{2}$
C. $\frac{5}{4}$
D. $\frac{15}{4}$

22. The graph of $x \geq -3$ in *xy*-coordinate system is

A.


B.


C.


D.


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23. Which of the following points is NOT in the half plane of $2x - 3y \leq 9$?

A. (0, 4)
B. (4, 0)
C. (– 4, 0)
D. (0, – 4)

24. The equation of the circle having centre (0, – a) and radius 9 is

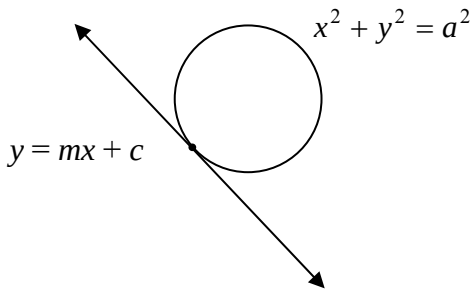
A. $x^2 + (y + a)^2 = 9$
B. $(x + a)^2 + y^2 = 9$
C. $x^2 + (y + a)^2 = 81$
D. $(x + a)^2 + y^2 = 81$

25. For the circle $x^2 + y^2 + 2gx + 2fy + c = 0$, if the centre is (0, 0) and radius is 9, then c is equal to

A. 81
B. 9
C. – 9
D. – 81

26. If $y = mx + c$ is tangent to the circle $x^2 + y^2 = a^2$, then c is equal to

A. $\pm a^2\sqrt{1 - m^2}$
B. $\pm a^2\sqrt{1 + m^2}$
C. $\pm a\sqrt{1 - m^2}$
D. $\pm a\sqrt{1 + m^2}$



27. The vertex of the parabola $(y + 3)^2 = 8(x - 1)$ is

A. (1, – 3)
B. (– 1, 3)
C. (3, – 1)
D. (– 3, 1)

28. The axis of the parabola $(x + 1)^2 = 6(y - 2)$ is

A. $x = 1$
B. $y = 2$
C. $x = -1$
D. $y = -2$

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For the ellipse $\frac{x^2}{81} + \frac{y^2}{9} = 1$, answer Q.29 and Q.30.

29. The foci are

A. $\left(\pm 6\sqrt{2},\ 0\right)$
B. $\left(0,\ \pm 6\sqrt{2}\right)$
C. $\left(\pm 3\sqrt{10},\ 0\right)$
D. $\left(0,\ \pm 13\sqrt{10}\right)$

30. The length of the minor axis is

A. 81
B. 18
C. 9
D. 6

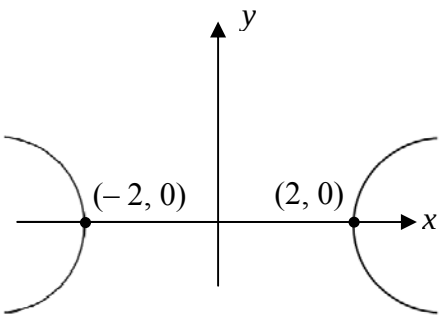
31. The transverse axis of the conic $5x^2 - 4y^2 = 20$ is

A. the *x*-axis.
B. the *y*-axis.
C. parallel to *x*-axis.
D. parallel to *y*-axis.

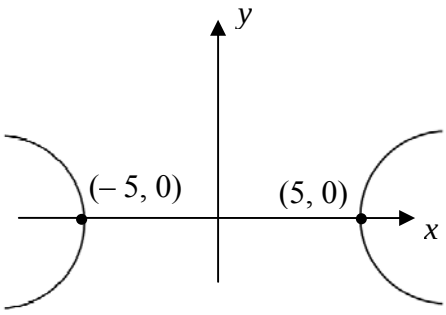
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32. Which of the following graphs represents the hyperbola $\frac{y^2}{4} - \frac{x^2}{25} = 1$?

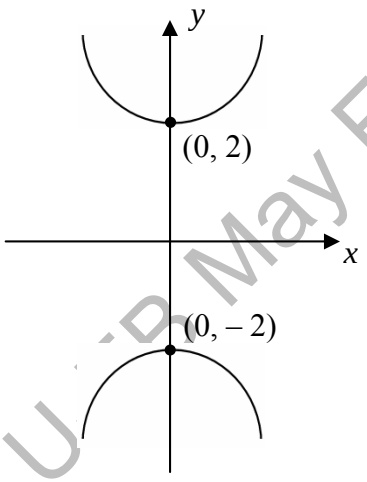
A.



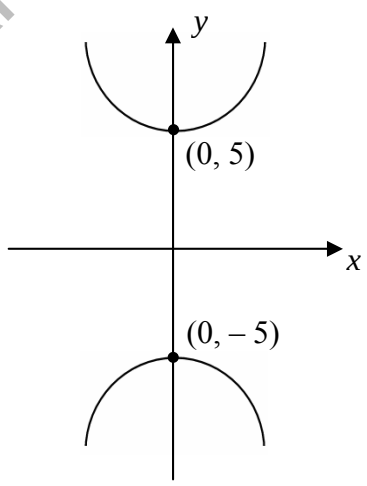
B.



C.



D.



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33. The equation of an ellipse with vertices $(\pm 7, 0)$ and co-vertices $(0, \pm 3)$ is

A. $\frac{x^2}{3} + \frac{y^2}{7} = 1$

B. $\frac{x^2}{7} + \frac{y^2}{3} = 1$

C. $\frac{x^2}{9} + \frac{y^2}{49} = 1$

D. $\frac{x^2}{49} + \frac{y^2}{9} = 1$

34. The coordinates of point *P* are (– 4, 12). If the axes is translated through the point O', the new coordinates of *P* are (3, – 5). The coordinates of O' are

A. (– 7, 17)

B. (7, – 17)

C. (– 1, 7)

D. (1, – 7)

35. The coordinates of point *P* are (5, 7). If the axes is rotated anticlockwise about the origin through an angle of 60º, the *x*-coordinate of the new coordinate of point *P* is

A. 5 cos 60º + 7 sin 60º
B. 5 cos 60º – 7 sin 60º
C. 5 sin 60º + 7 cos 60º
D. 5 sin 60º – 7 cos 60º

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